Scenario Analysis: Interest Rate Shocks

Part E(iv) — MBS Prepayment under Macro Stress

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TV Cox is the only model with a direct rate channel for scenario analysis.

Why TV Cox (E(ii)) is the primary model

`rate_incentive` $= r_{\text{orig}} - r_{\text{market}}(t)$ is a **direct time-varying covariate**.

A Δr bp market rate shock immediately shifts the incentive by $-\Delta r/100$ pp, giving an analytic hazard response:

$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}$$

Output: calibrated $S(t)$, CPR, WAL, and price — all in realistic ranges.

Why static DeepCox cannot do this

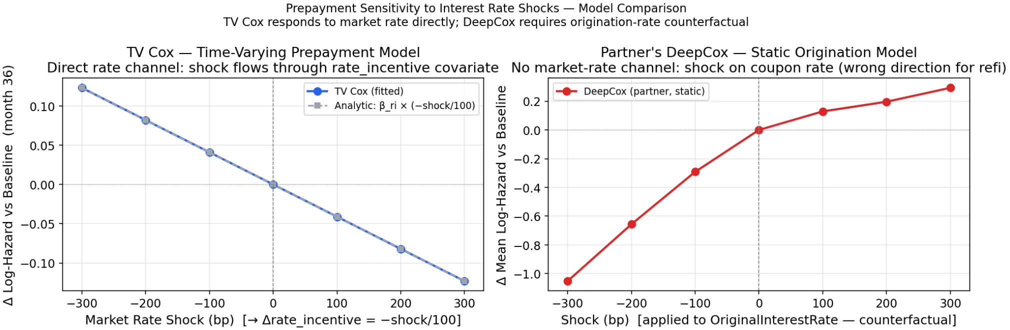
Static models (Part D) use `OriginalInterestRate` — the **coupon**, not the market rate.

Shocking this field gives the **wrong economic direction**: a rate *cut* reduces the coupon $\Rightarrow$ model predicts *less* prepayment (opposite to reality).

Static models require architectural changes before scenario analysis is valid.

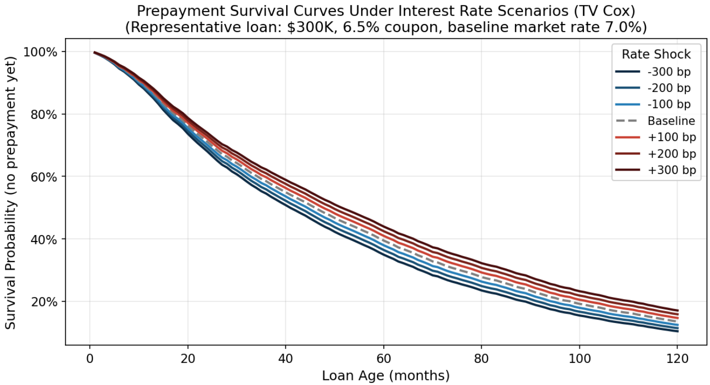
Representative loan: \$300K, 6.5% coupon, baseline market rate 7.0%.
Scenarios: ± 100, ± 200, ± 300 bp — covering the BCBS standard stress range.

TV Cox tracks the analytic formula exactly; DeepCox sign is reversed.



Left: TV Cox fitted $\Delta \log h$ overlaps the analytic line perfectly ($\hat{\beta}_{\mathrm{ri}} = 0.041$, HR $= 1.042$ per 100 bp).
Right: DeepCox $\Delta \log h$ is *opposite sign* — a 300 bp cut produces -1.05 (less prepayment), not more.

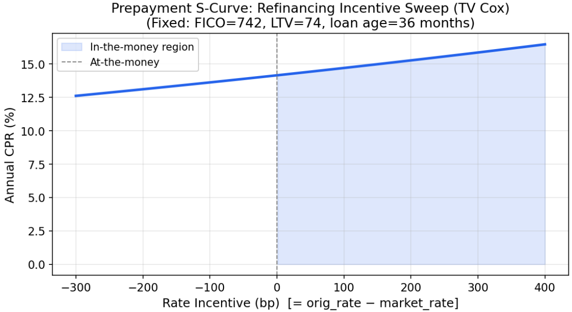
Rate scenarios shift survival by 6.7 pp and median prepayment by 11 months.



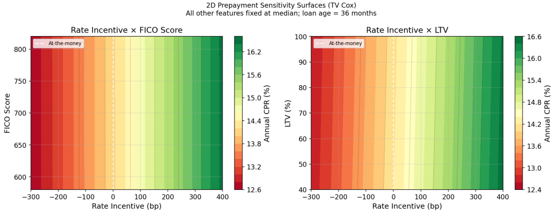
Scenario	10-yr	Med. mo.
−300 bp	10.4%	41
−200 bp	11.4%	43
−100 bp	12.5%	45
Baseline	13.5%	46
+100 bp	14.7%	48
+200 bp	15.8%	50
+300 bp	17.1%	52

Curves monotonically ordered.
±300 bp: 6.7 pp survival range,
11-month shift in median prepay.

The S-curve is smooth; FICO gates refi access, LTV caps the upside.

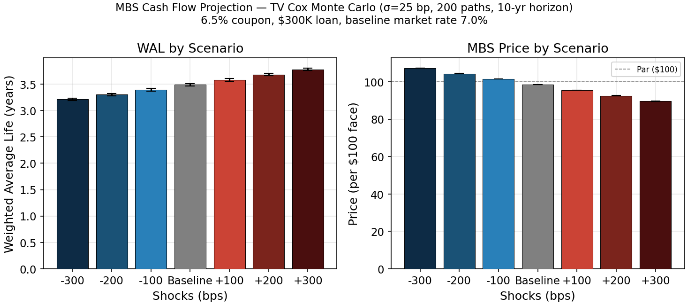


CPR: 12.6% at $-300\,\mathrm{bp} \rightarrow 16.5\%$ at $+400\,\mathrm{bp}$.
Flat slope reflects compressed $\hat{\beta}_{\mathrm{ri}} = 0.041$
(penalizer + 500K-row subsample).



FICO > 720: credit access enables refinancing.
LTV > 90: equity constraint flattens CPR regardless of incentive.

WAL range 0.57 yr; price range \$17.70 — negative convexity confirmed.



Scenario	WAL	Price (\$)
-300 bp	3.21	107.35
-200 bp	3.30	104.42
-100 bp	3.39	101.48
Baseline	3.48	98.52
$+100$ bp	3.58	95.56
$+200$ bp	3.68	92.60
$+300$ bp	3.78	89.65

MC: 200 paths, $\sigma = 25$ bp.
Base $<$ par: rate $>$ coupon.

-300 bp gain (\$8.83) $<$ $+300$ bp loss (\$8.87):
negative convexity confirmed.

Key Takeaways

1. **Time-varying covariates are required for rate scenario analysis**: static models produce the wrong sign when the shock variable is a coupon rate, not a market rate.
2. **TV Cox tracks the analytic formula exactly**: $\Delta \log h = \hat{\beta}_{\text{ri}} \times (-\Delta r/100)$ with $\hat{\beta}_{\text{ri}} = 0.041$ matches fitted values at all seven shock levels.
3. **Rate sensitivity is currently compressed**: WAL range 0.57 yr vs. 2–10 yr for agency MBS. Root cause: 500K-row subsample (0.7% of panel) + Ridge penalizer. Lower penalizer and larger sample are the two highest-leverage fixes.
4. **Negative convexity is captured**: price upside (\$8.83) $<$ downside (\$8.87) at $\pm$300 bp, consistent with prepayment curtailing price appreciation.
5. **FICO gates access; LTV caps the ceiling**: 2D contour confirms both constraints from E(i) and E(ii) coefficient signs.

Thank you

Questions?

Andersen & Gill (1982) *Ann. Stat.* · Sadhwani et al. (2021) *J. Fin. Econometrics* · BCBS (2019) *Interest Rate Risk in the Banking Book*